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Determining a reasonable function-evaluation budget for Part 1 (Randomized Optimization) can indeed be challenging, especially when trying to ensure comparability with the gradient-evaluation budget used in Part 2. Here are some guidelines and considerations to help you set an appropriate budget:

- **Understand the Nature of RO:** Randomized Optimization (RO) methods like Randomized Hill Climbing (RHC), Simulated Annealing (SA), and Genetic Algorithm (GA) are typically more computationally expensive per evaluation compared to gradient-based methods. This is because each function evaluation involves computing the full validation loss, which can be costly.
- **Magnitude of Budget:** A good starting point is to consider the scale of your problem. If your neural network's last 1-3 layers have around 50,000 parameters, you might start with a budget in the range of thousands of function evaluations. This is because RO methods need a sufficient number of evaluations to explore the parameter space effectively.
- **Comparability with Part 2:** To ensure comparability with the gradient-evaluation budget in Part 2, consider the total compute time or wall-clock time as a common ground. If your gradient-based methods in Part 2 take a certain amount of time to reach a threshold, aim for a similar time frame for your RO methods. This might mean adjusting the number of function evaluations based on how long each evaluation takes.
- **Pilot Runs:** Conduct pilot runs with a smaller budget to gauge the performance and convergence behavior of each RO method. This can help you refine the budget before committing to full-scale experiments.
- **Plateau Rule:** Implement a plateau stopping rule where you terminate the RO run if there is no improvement after a certain number of evaluations. This can prevent unnecessary evaluations and help you stay within a reasonable budget.
- **Consistency Across Algorithms:** Ensure that the function-evaluation budget is consistent across RHC, SA, and GA. This consistency is crucial for fair comparisons.
- **Documentation and Justification:** Clearly document your chosen budget and provide a rationale for it in your report. Explain how you ensured comparability with Part 2 and any adjustments made based on pilot runs or other considerations.

> Sidebar

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